

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/324018661>

Letter Frequency Analysis of Languages Using Latin Alphabet

Article in *International Linguistics Research* · March 2018

DOI: 10.30560/ilrv.1n1p18

CITATIONS

19

READS

4,197

2 authors:



Gintautas Grigas

Vilnius University

35 PUBLICATIONS 69 CITATIONS

[SEE PROFILE](#)



Anita Juskeviciene

Vilnius University

28 PUBLICATIONS 481 CITATIONS

[SEE PROFILE](#)

Letter Frequency Analysis of Languages Using Latin Alphabet

Gintautas Grigas¹ & Anita Juškevičienė¹

¹ Institute of Data Science and Digital Technologies, Vilnius University, Lithuania

Correspondence: Anita Juškevičienė, Institute of Data Science and Digital Technologies, Vilnius University, Akademijos str. 4, LT-08663, Vilnius, Lithuania. Tel: 370-5210-9314. E-mail: gintautas.grigas@mii.vu.lt, anita.juskeviciene@mii.vu.lt

Received: February 23, 2018; Accepted: March 8, 2018; Published: March 26, 2018

Abstract

The evaluation of the peculiarities of alphabets, particularly the frequency of letters is essential when designing keyboards, analysing texts, designing alphabet-based games, and doing some text mining. Thus, it is important to determine what might be useful for designers of text input tools, and of other technologies related to sets of letters. Knowledge of common features among different languages gives an opportunity to take advantage of the experience of other languages. Nowadays an increasing amount of texts is published on the Internet. In order to adequately compare the frequencies of letters in different languages used in the online space, Wikipedia texts have been selected as a source material for investigation. This paper presents the Method of the Adjacent Letter Frequency Differences in the frequency line, which helps to evaluate frequency breakpoints. This is a uniform evaluation criterion for 25 main languages using Latin script in order to highlight the similarities and differences among them. Research focuses on the letter frequency analysis in the area of rarely used native letters and frequently used foreign letters in a particular language. The frequency of the letters is one of the factors that determines the location of the keys for the language specific letters on the keyboard.

Keywords: diacritics, keyboard layout, Latin script, letter frequency, language statistics, language similarity

1. Introduction

Latin script is used in many languages but alphabets of these languages differ. A certain uniformity is ensured by the Basic Latin alphabet, which is actually the English alphabet with 26 letters. It can be regarded as the backbone of the alphabets of all languages that use Latin script. However, some letters of the Basic Latin are not included in the alphabets of other languages (Dagienė et al. 2010). On the other hand, every language has some specific letters. Thus, the variety of Latin-based alphabets is high and far higher than of other script groups (e.g. Greek, Cyrillic).

It is important to evaluate peculiarities of alphabets, particularly the frequency of letters, when designing keyboards, analysing texts, designing alphabet-based games, and performing other text-based tasks. This is especially important in the case of on-screen keyboards. They are included into the software, and thus, it is possible to select the optimal number of keys for each language, depending on the frequency of letters and the screen size restrictions that are common for mobile devices.

Another issue is the layout of the characters on the keyboard. The selection of the optimal layout, or at least a layout close to the optimal one, requires a considerable amount of research. On the basis of such a study (Bi et al. 2012), the English on-screen keyboard was simultaneously optimized for five languages: English, German, French, Spanish, and Chinese (pinyin). Although the five languages studied in this work represent a large population of potential users, one would still want to ask whether the result of such optimization can be further extended towards a number of other languages and their similarity groups.

Typing for mobile devices has some peculiarities due to their small size and holding on hands while typing. A model of two-thumb text entry was developed (MacKenzie, 2002). Keyboard layout QWERTY was adopted to this method by splitting it in two parts: one for the left hand, another for the right hand. Solution was simple but typing was slow. A special layout for two-thumb text typing has been designed by (Oulasvirta et al. 2013) and named KALQ. However, for English language only.

Similar investigation using bigrams of letters was made by Chun (2015). In referring to KALQ app for a certain type of smart phones he has indicated that this layout allows a 34% increase in typing speed. As a result of research Chun has developed two-thumb text entry keyboard layout for Korean language.

Conduction of similar studies for each of the languages is a difficult task. Could the research results of any other language be used? For this purpose, the frequencies of alphabet letters of different languages will be compared and similarities as well as differences among languages will be disclosed.

Our goal is to analyse the alphabets of various languages and to compare the letter frequencies of different languages by focusing on language specific letters and frequently used foreign letters. A similar research has been conducted for Lithuanian alphabet letter frequency comparison with the alphabets of other European languages (Grigas & Juškevičienė, 2015). In this paper, the investigation is expanded to 25 languages.

The number of language specific letters is usually lesser than of those taken from the Basic Latin alphabet (Dagienė et al. 2010) and their frequencies are usually lower. In order to reduce the keyboard size of mobile phones, tablet computers and other small devices, some language specific letters are left without keys. Thus, their typing becomes complicated: usually requiring to press (touch) more than one key. Language specific letters or at least part of them are often left out. Nevertheless, the increasing globalisation requires foreign language letters that are not included in the alphabet. In order to specify a threshold below which the typing of letters becomes considerably slower, for example, when typing with a few keystrokes, the frequencies of rarely used letters should be known.

Frequencies of letters depend on the type of text (topic, purpose, etc.), as well as the style of its author. There are many sources of letter frequency statistics available, often several per language. However, their samples differ. Our analysis requires to look for larger sources covering the same type or at least similar types of texts.

Letter frequencies of Danish, English, Finnish, French, German, Icelandic, Polish, Russian, Spanish and Swedish languages are available on the Practical Cryptography (2015) website. The presented samples are large with no less than 90 million characters for each language. Unfortunately, only ten languages are presented on this website.

The character (including letters) frequency statistics of the texts of Wikipedia is provided by Denny Vrandečić on his website (Vrandečić 2012) and is additionally analysed in the paper Language Resources Extracted from Wikipedia (Vrandečić et al. 2011). The author provides numbers of unigrams (characters), bigrams, trigrams of 262 languages in Wikipedia. All letters, including all foreign letters (regardless of the language) are counted. These two unique features are essential for our research.

Wikipedia is developed by many authors. The topics of articles vary, but have a lot in common among all languages. Moreover, in Wikipedia global phenomena are described, thus foreign letters are frequently used in its texts revealing more evident similarities and differences between languages. In addition, the texts of Wikipedia are posted online. This feature corresponds to contemporary tendencies in text.

Thus, this source of data is well-suited for our research and for this reason we have selected it. In the article, we will call this source (Vrandečić 2012) a Base of Wikipedia characters or only a Base.

The paper is structured as follows. Section 2 presents the sets of letters. Section 3 covers the proposed Method of the adjacent letter frequency differences in the frequency line, which helps to evaluate frequency breakpoints in the subsequent section 4 for all 25 languages. The areas of large jumps in the letter frequency lines, in order to decide which letters are reasonably to include in the main plane of keyboard, are covered in Sections 5. Similarities (or differences) among alphabets of the languages using correlation method are described in Section 6. The paper ends up with conclusions.

2. The Sets of Letters

The European Union has 24 official languages and 22 of these languages use Latin script. All of them are included in the analysis. We have also included some additional languages that are commonly used in Europe: Icelandic, Norwegian and Turkish. Thus, a total of 25 languages have been analysed. Statistics of all these languages is presented in the Base.

All characters, not only letters, are included in the Base. However, only letters of alphabets from the selected 25 languages are important for our analysis. Accordingly, the alphabets of these languages have been joined into one common set of letters. Everson (2004) has collected a lot of information about many alphabets. Alphabets of European languages are defined by ETSI in ETSI ES 202 130 v.2.1.2 (ETSI, 2007) standard and the annex of ISO 12199 (ISO 2000) standard. As the annex is of informative nature, the priority is given to the ETSI standard. Further on we mention only these two standards, so no ambiguity will arise if we call them only as ETSI and ISO.

In ETSI, letters of every language are divided into two groups: A and B. Group A includes all letters of the alphabet. They are compulsory. The letters that may be necessary for borrowed words or foreign personal names are assigned to group B. Nevertheless, this division has some doubtful points. For example, group A of the English alphabet includes

a few letters that are not in the Basic Latin alphabet (e.g.: æ, ñ). However, such letters usually belong to some other alphabet (e.g.: æ – Danish, ñ – Spanish) and thus they will be included into the joint letter set through those languages. The letter frequencies can reveal the real picture of the letter usage. Our preliminary study of the frequencies showed that two letters of the Latvian language are assigned to group A in ETSI, but their frequencies in Latvian are very low (ō – 0.0005% ģ – 0.0003%). In addition, they are absent in all other languages. Therefore, these two letters have not been included into the joint letter set.

There are less problems with letters from group B. If some letters are unreasonably moved into group B, they will be included into joint set from some other language.

Thus, the set of 102 letters has been formed:

aaaaaaãäæbcccçddđðëèéêëëçfgggghhiijlklmnnñnoóòôõöøœpqrřrssşßtţpuúûüũüüvwxyýzžžž

All of the other characters (i.e. not letters) have been removed from the Base. Accordingly, the calculation calculation samples of different languages letter frequencies have been developed (Table 1).

Table 1. The amount of the analysed texts

Language code	Language name	Text amount (in millions of letters)
cs	Czech	294
da	Danish	170
de	German	820
en	English	806
es	Spanish	787
et	Estonian	84
fi	Finnish	349
fr	French	763
ga	Irish	12
hr	Croatian	132
hu	Hungarian	358
it	Italian	784
is	Icelandic	24
lt	Lithuanian	103
lv	Latvian	43
mt	Maltese	53
nl	Netherlands	734
no	Norwegian	297
pl	Polish	714
pt	Portuguese	672
ro	Romanian	173
se	Swedish	388
sl	Slovak	109
sk	Slovenian	90
tr	Turkish	216

3. Ratios, Differences and Breakpoints of Letter Frequencies

It is common to express letter frequencies as percentages. Two additional attributes for the letter pair analysis will be considered: the ratio r of the frequency f and the relative difference d of the letter pair. The ratio $r(i, j) = f_j/f_i$ shows how many times the frequency of letters reduces in the descending order line of the frequencies, where $i < j$.

In the most cases we will be interested in the ratio of the adjacent letters, that is, when $j = i+1$. The relative difference of frequencies $d(i, j) = (f_i - f_j)/f_j = f_i/f_j - 1$ will be called the difference.

The difference shows how many times the frequency f_i of the letter i is higher than the frequency f_j of the letter j (or vice versa). The difference of the adjacent letters will be 0 if the frequencies of both letters are equal or greater than 0 if the letters are sorted by the descending order of their frequencies. In this case the difference is more convenient to use than the ratio.

The large differences of the adjacent letters will be indicated as the breakpoints of frequencies. The breakpoints of frequencies, which are greater than 1 (frequency of the next letter is reduced two or more times) will be identified.

The English language will be used as an example. Table 2 contains letter frequencies and differences (presented in four decimals until the frequency value has at least two significant digits). Only 20% of rarely used characters are not included.

Table 2. Letter frequencies and differences of the English language

Letter	Frequency	Difference
e	12.15476703	0.401597
a	8.672083734	0.008609
t	8.598064819	0.141554
i	7.53189498	0.020975
o	7.377156799	0.005413
n	7.337440171	0.106191
s	6.633069973	0.001225
r	6.624957169	0.396094
h	4.745351843	0.119989
l	4.236963167	0.085339
d	3.90381717	0.140023
c	3.424333016	0.272688
u	2.690630195	0.016587
m	2.646728388	0.149707
f	2.302089594	0.083845
p	2.124003257	0.09196
g	1.945129235	0.16274
w	1.672883441	0.078678
y	1.550863958	0.044312
b	1.48505829	0.399725
v	1.060964146	0.767008
k	0.600429726	1.974128
x	0.201884315	0.079297
j	0.187051741	0.412901
z	0.13238847	0.204144
q	0.109944054	8.833333
é	0.011180751	1.473379
á	0.004520435	0.421285
ö	0.003180528	0.155275
ü	0.002753048	0.016871
í	0.002707371	0.015173
ó	0.002666907	0.315818
ā	0.002026804	0.091511
ä	0.00185688	0.207523
è	0.00153776	0.28757
ø	0.001194311	0.168853
ç	0.001021781	0.098332
ñ	0.000930302	0.149893
š	0.000809034	0.025649

æ	0.000788802	0.017451
ú	0.000775272	0.066234
ł	0.000727113	0.028802
ū	0.000706756	0.01047
å	0.000699433	0.134488
ć	0.000616519	0.016994
â	0.000606217	0.013909
ë	0.0005979	0.114272
à	0.000536584	0.032728
ã	0.000519579	0.020229
ī	0.000509277	0.049898
č	0.000485073	0.025722
ô	0.000472909	0.184333
ǒ	0.000399304	0.218099
ǎ	0.000327809	0.040173
ş	0.000315148	0.011151
ê	0.000311673	0.050188
ē	0.000296778	0.022669
ś	0.0002902	0.037267
ž	0.000279773	0.101662
ß	0.000253956	0.039106
ı	0.000244398	0.049014
î	0.000232979	0.005895
ń	0.000231614	0.019115
ď	0.000227269	0.077059
ï	0.000211009	0.370968
ò	0.000153913	0.036789
ì	0.000148451	0.065004
þ	0.00013939	0.158927
ý	0.000120275	0.004145
û	0.000119779	0.072222
ǵ	0.000111711	0.026226
ř	0.000108856	0.013873
ę	0.000107366	0.075871
ş	0.0000998	0.166909
ž	0.0000855	0.004373
ą	0.0000851	0.007342
ù	0.0000845	0.022523
ě	0.0000827	0.045526
ö	0.0000791	0.00315
œ	0.0000788	0.480186
õ	0.0000532	0.043796
è	0.0000510	0.374582

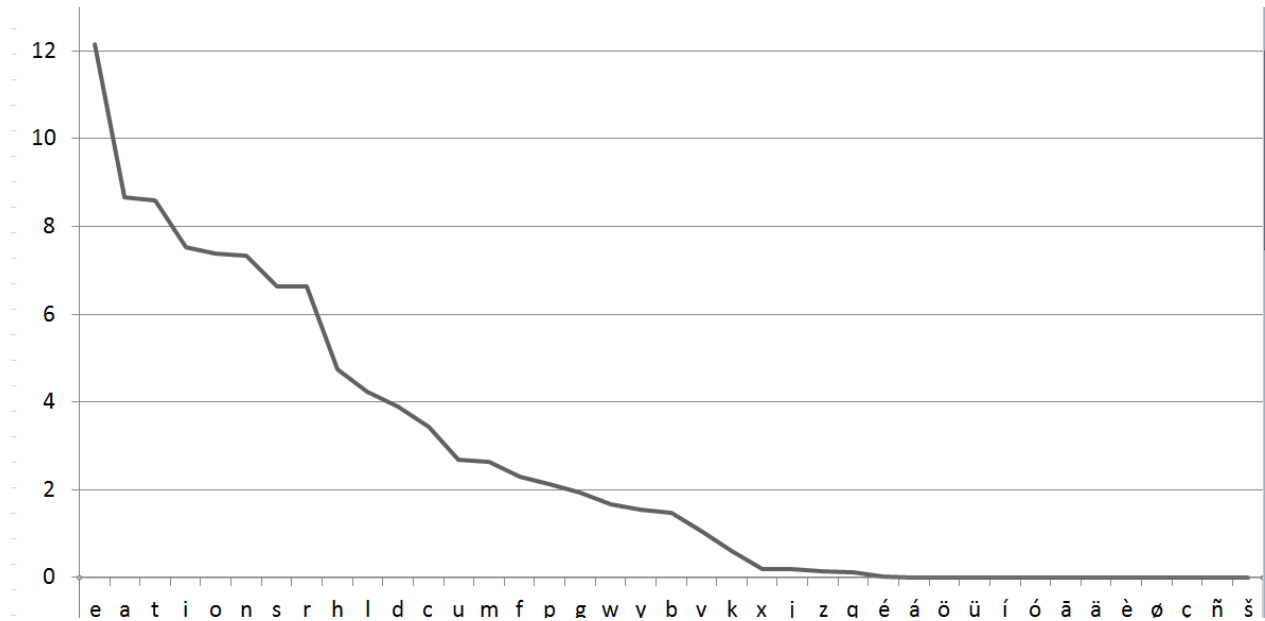


Figure 1. Letter frequencies of the English language in percent

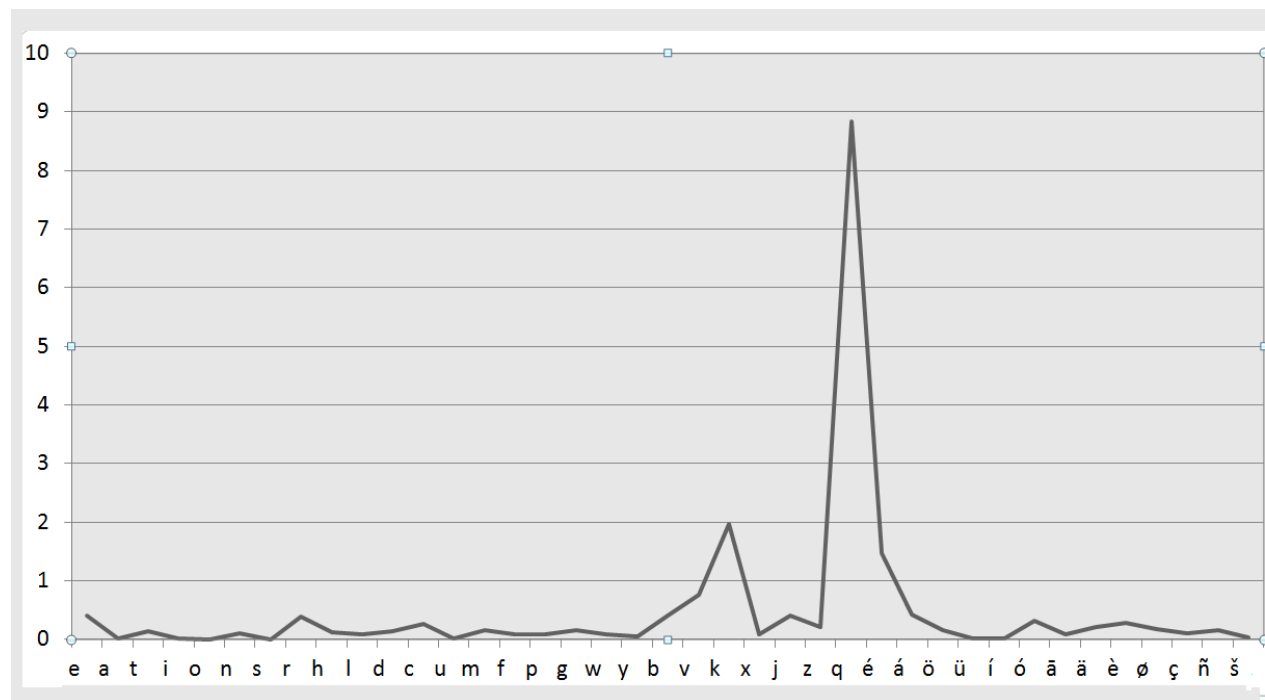


Figure 2. Letter frequency differences of the English language

The letter frequencies are presented in Figure 1, whereas the differences are introduced in Figure 2.

Two jumps can be seen in the differences chart: a small one (1) between the letters k and x, and another one, a very large one (8), between the letters q and é that clearly draws the line between the letters belonging to the English alphabet and those that do not belong to it or, in other words, between the commonly (often) used and rarely used letters. This threshold is important when designing data input devices (keyboards).

The use of the letters of the alphabet is balanced and the alphabet corresponds to the actual usage of the letters in the online space if all of the alphabet letters of the used language are listed before the first larger jump and all foreign language alphabet letters take the position after the jump.

There are some borrowed words in English texts that contain accented characters, such as à la carte, abbé, Ægean, archæology, belovèd, café. According to Everson (2004) the English alphabet contains 16 extra letters (à, æ, ç, ð, é, è, ê, ë, ì, ñ, ò, ô, œ, 3, þ, þ). However, as it can be seen in Figure 2, these letters are positioned after the high jump (8). Therefore, their position in the frequency line shows that they must be left outside the English alphabet.

4. Frequency Differences and Jumps Between Adjacent Letters in the European Languages

The frequency differences are introduced in a simplified form, as the presentation of a separate chart for every language (Table 3) would occupy a lot of space. The letters are divided into 4 groups of frequencies and are set out in the descending order. The most common letter is first in the line. It can be seen that the most common letters are: e (in 12 languages) and a (in 10 languages).

Table 3. The letters of 25 languages presented in their frequency descending order with inserted jump indicators

	$f \geq 0.1$	$0.1 > f \geq 0.01$	$0.01 > f \geq 0.001$	$0.001 > f \geq 0.0001$
cs	oeantsilvrkdumpíchzáyjběěřýčžšgůf 1 úw	xñ	ót 1 d'q	üöä 1 čèlèl'
da	erntaisldlogmkfvubhpâyøæcj 2 w	z	xéqöü	ä 1 áoíèðäšßlç
de	enirstadhulgocmbfkwpvüäöjyß	1 x	1 q 1 é	1 áoíèšççäløâcúâüð
en	eationsrhldcumfpgwybv 1 xjq	8 é	1 áoüíóääèøç	ñšæ...èà...ê...î...æð
es	eaonsrildtcmpbgvfýóhqjzázéñkú	w	1 ü 1 øèçàää	ääüèšçðò
et	aiestlunokrdmvgpjhäbõü 1 fçö 1 y	wz	šx 1 žqé	ááóoía
fi	aintesloukämrvjhpyd 1 ögcbfw	z	x 1 šéq	1 äüäóóí 1 èøç
fr	easnitrluodcpmévfgbhqàxèyjkêz	wçôâîüæüü	2 éáuüíóääüñ	šçüä
ga	ainhrestclodgmubíáéíóú 4 vyk	wjzqxq	1 äðü 1 àlèèçñòšäç	äðð...æ...ê...qí...æ
hr	aioenrjstuklvdmpgzbcñšžćfđyw	2 x 1 qé	áuüíóáuääçñäè	
hu	eatlsnkriozáégmbyvdhupjőfcóóüüü	1 wx	3 q	2 šäðçčäçšçžlø
is	arniestulgmðkfovhdiájðóbypðæúéý	1 wx	z 2 q	üöäääèšçlāīāūā
it	eiaonltrscdupmgvfbzh 1 qèàkyò	wüjxié	8 áoüüíóäçšçéäüñā	üð
lt	iasoret nukmlpvdjgēbyųšžcaįčūfzhę	1 x	w	5 qéáoíóüāīāñ
lv	aisetrunāoklmdpvjzīgēčš 1 ņflūhžķģčl	1 ywx	2 q 1 éäüð	óí...ð...ŗ
mt	ailtenrumsojkdfbhpqwhzğžxqvčç 2 y	à 4 éü	òèäüçíóðçìšçā	
nl	enairtodslghvmukcpbwjzf 2 y	1 xèé	1 qèüüð	óáāīlçšðçāāāā
no	erntasioldgkmvfpubhåyjøc 1 æw	zxéqöä	üáoíèäðäüāšlçtīßüçð	ñè
pl	aioenrzwscdkypmuljlgbhąęóźśf 1 ńcy	žx 2 éq	áuð	íäšèçø
pt	aeosirdntmuclp 1 gvbfhãqéçázíxókyêwõú	ääð	4 ü 1 ðèäñlş	ääšè
ro	eiarntulocsdpmăřvīgbsţzhâjxkyw	4 éq	áuðèóíáðäşçşlúcç	ßāāóè
sk	oaienrvtslkdpuchjbzáyýíčéšžúgfł'ťóôwx	ñď'ä	4 íq	řčöřèü 1 řüí
sl	eaionsrltjvkdpmuzbgħčšžf 2 y	w 2 xéqčá	üðóíèäüðçäððóøł	èäāñè
sv	eanrtsildomkgvhfupäcböåyj 3 xw	zé 1 q	1 üäøèóíæðäçšçèçú	ðñlā
tr	aeinrlkldtsmyuöüšvgzhpcğöf 4 j	w 2 xâîq	é 1 üáoíäæš	ñçúèäç

Language specific letters are bolded in group A of the ETSI standard (Table 3). The letters of Basic Latin alphabet, which ETSI assigned to group B, are underlined. The integer value of the differences is written in white on a black background and it is inserted in the breakpoint place.

The biggest jump (11), marked by a square, belongs to the Spanish language. English and Italian languages also have quite big jumps (8). Next is the Lithuanian language (5). The bigger the jump, the higher the disjuncture between often and rarely used letters (letter groups A and B according to ETSI).

All languages sorted in descending order by their biggest jumps (J) and the distances from the biggest jump to the nearest smaller jump (D) are presented in Table 4.

If a different gradation of jumps was chosen (e.g., 0.5, 1.5, 2), a slightly different language distribution into the groups would be achieved. However, languages would not move more than two positions in the table (after the appointment of the position for each of the two criteria).

Due to the fact that there is no language that would not have any breakpoints, and values of these breakpoints vary in a wide range (from 1 to 11), it can be assumed that the given gradation by integer 1 is appropriate for our purposes.

Table 4. Jumps of the letter frequencies

J	D	Languages
11	10	es
8	7	en, it
5	4	lt
4	4	ro
4	3	ga, pt, sk
4	2	mt, tr
3	2	sv
3	1	hu
2	2	fr
2	1	da, hr, is, lv, nl, pl
2	0	sl
1	1	no
1	0	cs, de, et, fi

Adding the language specific letters into the ETSI standard raised objections inside it. The letters of the main group A should be treated and typed as the other letters of this group and the letters of the group B are moved further away, after numerals, and thus require more keystrokes to type them. Unfortunately, this rule applies only to the English language, which alphabet has only 26 letters of the Basic Latin. Other language specific letters within group A are typed in the same way as the letters of group B.

It can be assumed that a contradiction appeared in the ETSI standard due to the layout of the frequency unjustified letters (layout does not meet the distinction of the letters between the groups A and B) and it resulted in the decreased SMS typing speed in languages other than English.

Jumps of the frequencies can be used when designing onscreen keyboards. The number of keys is not mechanically restricted. Therefore, an optimal number of keys can be selected for each language by splitting the list of letters at the most appropriate breakpoint.

Table 5. The ratio of frequencies

Language	Language specific letters				Last in the en		Ratio of freq.	
	List	f_{s1}	f_{sn}	f_{sall}	Letter	f_{bn}	f_{sn}/f_{bn}	f_{s1}/f_{bn}
cs	íáěřýčžšůňót'd'	2.7952	0.0170	12.5090	q	0.0120	1	233
da	åøæ	0.7926	0.7320	2.2866	q	0.0154	48	51
de	üäöß	0.5672	0.1567	1.5488	q	0.0344	5	16
es	óíáéñú(ü)	0.7998	0.1252	2.2796	w	0.0831	2	10
et	äõüö(šž)	1.0694	0.2330	2.9515	q	0.0117	20	91
fi	äö(å)	3.3433	0.4363	3.7796	q	0.0136	32	246
fr	éàèêçôâîùœü(ë)	2.4438	0.0151	3.6890	w	0.0881	0	28
ga	íáéóú	1.8851	0.8751	7.1006	q	0.0124	71	152

hr	čšžćđ	0.8837	0.1981	2.7016	q	0.0154	13	57
hu	áéőóíüúű	3.5430	0.2253	11.3774	q	0.0122	18	290
is	ðíáóþöæúéý	3.6600	0.2497	12.0487	q	0.0109	23	336
it	èàòùìé(ó)	0.2381	0.0411	0.7084	x	0.0585	1	4
lt	ėųšžąįčūę	1.6643	0.1721	6.8919	q	0.0054	32	308
lv	āīēšņļūžķģč(ōŗ)	4.0671	0.1169	10.2346	q	0.0064	18	635
mt	ħġżċ	2.0288	0.5814	4.1295	q	0.6844	1	3
pl	łąęóźśńćż	1.7848	0.0660	6.0278	q	0.0121	5	148
pt	ãéçáíóêôúââô(ü)	0.6669	0.0412	3.2523	w	0.1235	0	5
ro	ăîșțâ	2.2527	0.4485	5.7356	q	0.0156	29	144
sk	áýíčěšžúľťóňďäĺŕ	1.7484	0.0067	9.0332	q	0.0126	1	139
sl	čšž	1.1240	0.5336	2.5372	q	0.0132	40	85
sv	äöå	1.6661	1.2724	4.2160	q	0.0194	66	86
no	åøæ(é)	1.0216	0.1678	1.9692	q	0.0176	10	58
tr	ıüşçğö	8.5689	0.8869	10.5980	q	0.0124	72	691
Average		2.1136	0.3303	5.5394		0.5609	22	166

Note. English and Dutch languages which do not have the language specific letters are not included in the Table.

5. Frequency Ratio of Remote Letters

The ratio of the frequencies and their differences can be analysed not only between adjacent letters but also among remote letters. Such differences may be useful when considering the letter layout options on the keyboard, for example, what letter or group of letters should be allocated in the more convenient positions for typing..

Frequency differences of language specific and lesser-used Basic Latin letters are presented in Table 5. The following notations are used:

f_{s1} – the frequency of the most frequent (at the front of the list) language specific letter,

f_{sn} – the frequency of the rarest (at the end of the list, not in parenthesis) language specific letter,

f_{sall} – the sum of frequencies of all language specific letters,

f_{bn} – the frequency of the rarest Basic Latin alphabet letter,

f_{s1}/f_{bn} – the ratio of frequencies of the most frequent language specific letter and the rarest letter of the Basic Latin alphabet,

f_{sn}/f_{bn} – the ratio of frequencies of the rarest language specific letter and the rarest Basic Latin alphabet letter.

The letters in parenthesis are language specific letters that are included in group A of the ETSI standard. Nevertheless, they should not belong to this group due to their frequencies (see Table 5). Language specific letters are mostly used (more than 10%) by the Czech, Hungarian, Icelandic, Latvian and Turkish.

The ratio of the frequencies indicates the importance of the language specific letters and numeric evaluation of this importance with regard to rarely used (f_{sn}/f_{bn}) and commonly used (f_{s1}/f_{bn}) language specific letters. These ratios can be used to determine which letters should be given the priority when the data input devices (keyboards) have a limited number of keys.

These two columns of values can be considered as *from ... to* values. *From* – the number of times the frequency of the rarest language specific letter exceeds the frequency of the rarest Basic Latin letter. *To* – the number of times the frequency of the most frequent language specific letter exceeds the frequency of the rarest Basic Latin letter. Thus, if a language specific letter will be brought to the background of the keyboard, and a rarely used Basic Latin letter will remain in the foreground, this decision will be far away from the optimal option by *from* times.

For example, if the Latvian tablet computer keyboard foreground has only the Basic Latin letters, including rarely used q, and all the language specific letters are typed by few keystrokes, they are typed several times more slowly, even though that the letters č and ā are 18 and 635 times, respectively, more common than letter q. By the way, the frequency of letter q is the lowest among Basic Latin letters in 19 languages (from the 23 listed in the Table).

6. Character Frequency Correlation

The linguistic similarity among the characters in terms of the frequency can be judged from their alphabets' frequency correlations. Unigram frequency correlations of all the analysed language pairs are provided in Table 6.

In order to make Table 6 more compact, correlation coefficients (CC) are expressed as integer percentage that has been obtained by fractional CC multiplied by 100 and rounded to the nearest integer.

These percentage numbers range from 74 (da/ga) to 99: da/no (Danish/Norwegian) and hr/sl (Croatian/Slovenian). These two cases are depicted in frequency charts (Figure 3 and Figure 4). The letters are arranged in the decreasing order according to their frequency in the first language. As a result, the first language frequency polyline is permanently decreasing, whereas the second one is a zigzag curve. As it can be seen in the case of maximum correlation, the curves almost coincide, while in the case of minimum correlation the curves differ, although their correlation coefficient is not small.

Table 6. Unigram frequency correlation of languages as percentage

	cs	da	de	en	es	et	fi	fr	ga	hr	hu	is	it	lt	lv	mt	nl	no	pl	pt	ro	sk	sl	sv	tr
cs		85	82	88	90	87	87	87	76	92	88	81	90	86	83	82	86	86	86	88	85	98	94	89	82
da	85		96	94	91	85	80	94	74	84	87	85	89	79	79	81	97	99	79	87	87	84	88	96	84
de	82	96		94	90	85	81	95	79	82	82	83	89	77	77	80	97	95	79	86	90	80	85	93	83
en	88	94	94		95	91	89	95	86	88	87	86	95	86	84	87	94	94	87	93	93	89	90	96	85
es	90	90	90	95		91	86	97	85	90	85	85	97	87	86	86	93	90	86	98	95	91	91	93	87
et	87	85	85	91	91		96	91	86	93	87	88	92	94	94	94	86	89	83	90	91	88	82	91	89
fi	87	80	81	89	87	96		86	83	91	85	86	90	93	90	92	81	85	83	84	88	88	91	90	87
fr	87	94	95	95	97	91	86		90	86	87	85	94	85	84	85	94	94	81	93	94	86	89	94	84
ga	78	74	79	86	85	86	83	80		83	77	85	87	82	82	86	78	76	77	84	87	80	81	84	83
hr	92	84	82	88	90	93	91	86	83		83	85	93	94	91	92	87	86	89	91	90	95	99	89	87
hu	88	87	82	87	85	87	85	87	77	83		84	85	79	83	83	86	90	80	82	81	87	87	90	84
is	81	85	83	86	85	88	86	85	85	85	84		85	86	86	88	80	86	76	82	86	82	85	90	86
it	90	89	89	95	97	92	90	94	87	93	85	85		91	87	92	91	89	88	95	97	91	94	92	88
lt	86	79	77	86	87	94	93	85	82	94	79	86	91		94	92	78	81	85	88	89	89	93	86	86
lv	83	79	77	84	86	94	90	84	82	91	83	86	87	94		90	78	82	81	86	88	86	90	87	85
mt	82	81	80	87	86	94	92	85	86	92	83	88	92	92	90		81	83	82	84	91	83	91	88	90
nl	86	97	97	94	93	86	81	94	78	87	86	80	91	78	78	81		97	82	90	90	86	90	93	84
no	86	99	95	94	90	89	85	94	76	86	90	86	89	81	82	83	97		80	87	87	86	90	96	85
pl	86	79	79	87	86	83	83	81	77	89	80	76	88	85	81	82	82	80		86	85	88	89	83	82
pt	88	87	86	93	98	90	84	93	84	91	82	82	95	88	86	84	90	87	86		93	91	91	91	83
ro	85	87	90	93	95	91	88	94	87	90	81	86	97	89	88	91	90	87	85	93		86	90	91	88
sk	98	84	80	89	91	88	88	86	80	95	87	82	91	89	86	83	86	86	88	91	86		96	89	83
sl	94	88	85	90	91	92	91	89	81	99	87	85	94	93	89	91	90	90	89	91	90	96		92	87
sv	89	96	93	96	93	91	90	94	84	89	90	90	92	86	87	88	93	96	83	91	91	89	92		88
tr	82	84	83	85	87	89	87	84	83	87	84	86	88	86	85	90	84	85	82	83	88	83	87	88	

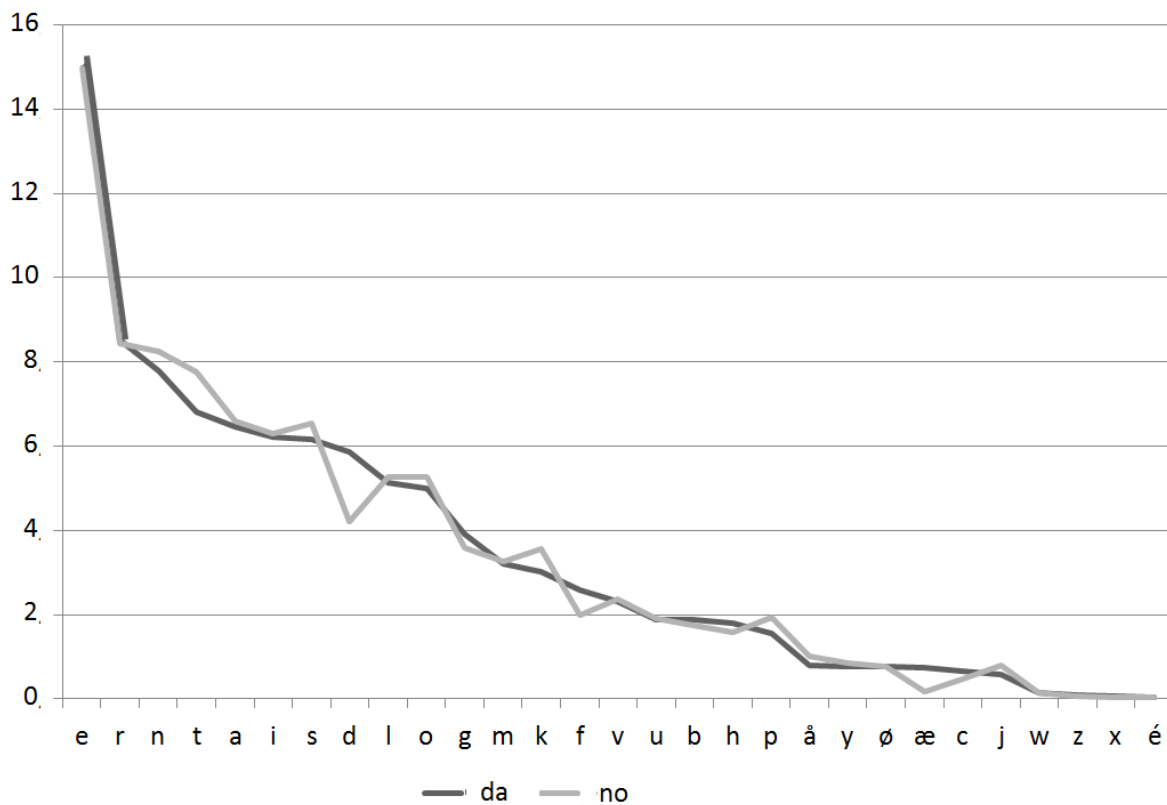


Figure 3. Comparison of letter frequencies in Danish and Norwegian languages

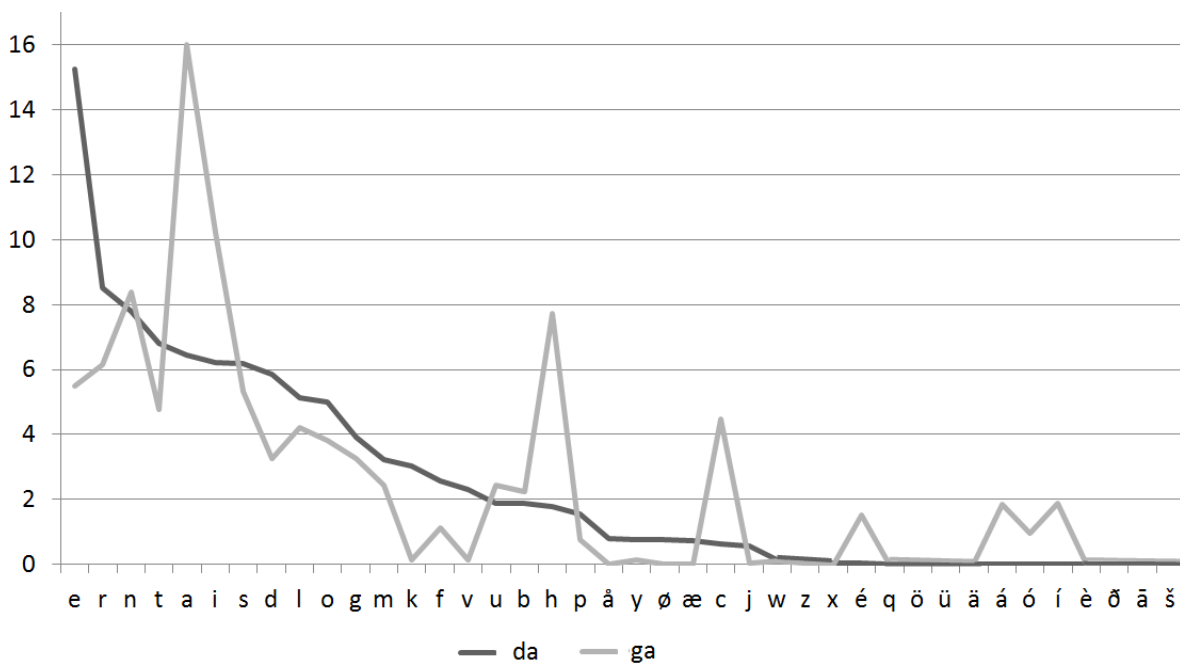


Figure 4. Comparison of Danish and Irish language unigrams

Strength of the correlation is usually described as follows:

CC = 0–0.2 very weak,

CC = 0.2–0.5 weak,

CC = 0.5–0.7 average,

CC = 0.7–1 strong.

According to this scale the correlations of all language pairs are strong. This is due to the similarity of the usage of letters in different languages. Therefore, in our case, it is reasonable to describe correlations ad hoc, with epithets only for limited range of correlations, for example:

CC = 0.74–0.80 very weak,

CC = 0.81–0.85 weak,

CC = 0.86–0.90 average,

CC = 0.91–0.95 strong,

CC = 0.96–0.99 very strong.

Let us list the language pairs with very strong mutual correlation:

CC = 99: da/no (Danish/Norwegian), hr/sl (Croatian/Slovenian);

CC = 98: cs/sk (Czech/Slovak), es/pt (Spanish/Portuguese);

CC = 97: es/fr (Spanish/French), es/it (Spanish/Italian);

CC = 96: da/de (Danish/German), et/fi (Estonian/Finnish), en/sv (English/Swedish), Slovak/Slovenian.

Spanish language belongs to three pairs of the strongest correlating languages, whereas Slovak, and Slovenian belong to two. By joining the Spanish language with all of its strongest correlation languages mentioned here, we can get a group of four languages: Spanish, Italian, Portuguese, French. An analogous group can be constituted with the Slovenian language: Slovenian, Croatian, Slovak.

This is a comparison of the alphabet letters by giving their frequencies corresponding weights. Therefore, the results should be more realistic and the optimisation of the letter layout on the keyboard may be based on them. The search for an optimal letter variant for every language by experiments is costly and time consuming. If the letter frequencies of the two languages are well correlated (CC is strong), it is likely that they will have similarities in their layout on the keyboard.

Most researches and practical works are conducted to establish a rational layout of the characters in the English language keyboard. The tablet's thumbnail keyboards are becoming relevant. KALQ layout for the English language is noteworthy (Bi et al. 2012). The English language correlates the best with the Swedish language (0.96). Therefore, Swedish has more opportunities to take advantage of what is done for English. If analogous works were done for Spanish, then three languages (Italian, Portuguese and French) could use their results. Similarly, two languages (Croatian and Slovak) could use the results obtained for Slovenian language.

7. Discussion and Conclusions

1. The letter frequencies are taken from a source common to all languages (Wikipedia), which ensures possibility for an adequate comparison of the letter frequency characteristics of all the official languages of the European Union and three additional languages (Icelandic, Norwegian and Turkish) employed in the online space that use Latin script. The common alphabet of all the analysed languages has been comprised of 102 letters and it has been included in the calculations of every language.

2. The original Method of the Adjacent Letter Frequency Differences is presented, which helps to highlight similarities and differences among languages. The results of this work can benefit screen-keyboard designers with the aim the number of keys to be minimal, which is important for mobile devices with small screens.

3. For all the analysed languages frequency jumps range from 1 to 11 that correspond to the frequency differences from 2 and 12 times between the two adjacent letters sorted by frequency. The biggest jumps are established in the area of rarely used letters. They divide the letters into two distinct groups: 1) commonly used and 2) rarely used. It is reasonable that the clear difference between the groups should be clearly reflected in the typing convenience and speed: the letters of the first group must be in the foreground of the keyboard and typed directly with one keystroke, whereas the second group may be in the background and typed by a few keystrokes.

4. There are some devices, especially mobile, which do not comply with the previous recommendations of these conclusions: some letters, usually the language specific ones, are located in the background layout and rarely used letters are left in the foreground. Such irrational solution is revealed by the ratio of the frequencies of those letters. This ratio depends on the particular letters and can be very high, it can reach even a few hundred times. Thus, it deviates from the optimal solution by so many times.

5. The maximum frequency jumps of particular language define the stability of the alphabet of that language. The largest is for Spanish (11), while the following languages are: English (8), Italian (8) and Lithuanian (5).

6. The correlation between the frequencies of all the analysed languages is calculated. The maximum correlation coefficient 0.99 is obtained from two pairs of languages: Danish – Norwegian and Croatian – Slovenian. The frequency of the letters is one of the factors that determines the location of the keys for the language specific letters on the keyboard. A high correlation coefficient indicates that the layout of the letters on those keyboards can be close to each other, and if the optimal layout of one language is established, then similar layout may be considered for the other language of the pair or at least may be used as a starting point for its layout optimisation.

7. Using the results of this work it is recommended to conduct researches with similar groups of languages identified or individual languages of those groups in order to optimize the layout of the keyboard for all group languages and the typing convenience (speed) for each language separately.

We believe that it would be useful to use the statistics of the bigrams.

References

- Bi, X., Smith, B. A., & Zhai, S. (2012). Multilingual touchscreen keyboard design and optimization. *Human-Computer Interaction*, 27(4), 352-382.
- Chun, C. (2015). A systematic technique of smart phone keyboard layout design for optimal text-messaging. *2015 IEEE International Conference on Consumer Electronics (ICCE)*, Las Vegas, 9-12 Jan. 2015. <https://doi.org/10.1109/ICCE.2015.7066545>
- Dagienė, V., Grigas, G., Jevsikova, T. (2010). *Programinės įrangos lokalizavimas* [Software Localisation. Monograph] // MII, 328 p.
- ETSI ES 202 130 v.2.1.2 (2007). Character repertoires, orderings and assignments to the 12-key telephone keypad (for European languages and other languages used in Europe).
- Everson, M. (2004). The Alphabets of Europe. Everttype. Retrieved April 20, 2017, from <http://www.everttype.com/alphabets/>
- Grigas, G., & Juškevičienė, A. (2015). Raidžių dažnių lietuvių ir kitose kalbose, vartojančiose lotyniškus rašmenis, analizė [Letter Frequency Analysis of Lithuanian and Other Languages Using the Latin Alphabet]. *Santalka: Filologija, Edukologija*, 23(2), 81-91.
- ISO 12199:2000 (2000). Alphabetical ordering of multilingual terminological and lexicographical data represented in the Latin alphabet.
- MacKenzie, I. S., & Soukoreff, R. W. (2002). A model of two-thumb text entry. *Proceedings of Graphics Interface, 2002*, 117-124.
- Oulasvirta, A., Reichel, A., Li, W., Zhang, Y., Bachynskyi, M., Vertanen, K., Kristensson, P. O. Improving two-thumb text entry on touchscreen devices. *CHI '13 Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, Paris, France - April 27 - May 02, 2013, p. 2765-2774. <https://doi.org/10.1145/2470654.2481383>
- Practical Cryptography (2017). Letter frequencies for various languages. Retrieved April 20, 2017, from <http://practicalcryptography.com/cryptanalysis/letter-frequencies-various-languages/>
- Vrandešić, D. (2012). Letter frequency. Retrieved April 20, 2017, from <http://simia.net/letters/unigrams.zip>
- Vrandešić, D.; Sorg, Philipp; Studier, Rudi (2011). Language Resources Extracted from Wikipedia. *Proceedings of the sixth international conference on Knowledge capture*, June 26-29, Baniff, Alberta, Canada, p. 153-160. <https://doi.org/10.1145/1999676.1999703>

Copyrights

Copyright for this article is retained by the author(s), with first publication rights granted to the journal.

This is an open-access article distributed under the terms and conditions of the Creative Commons Attribution license (<http://creativecommons.org/licenses/by/4.0/>).